

Half-Integer Quantization of Fermion Masses and the Universal Mass Ladder

A Transparent Analysis within the Yakushev Unified Coordination Theory

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(Corrected version with explicit resonance factors and improved hierarchy discussion)

Abstract

We present a corrected and transparent analysis of fermion masses and the universal mass ladder within the Yakushev Unified Coordination Theory (YUCT). The fundamental scaling quantum $q = (3/2)^{1/3} \approx 1.1447$ is used to show that all nine charged fermion masses are quantised in half-integer units of $\ln q$. The **raw half-integer rule** $m_{\text{base}} = m_e q^{N_f}$ predicts the masses with an accuracy of 2 – 5%. Small resonance corrections η_f (typically within $|\eta_f - 1| \lesssim 0.05$) further refine the predictions to sub-percent accuracy. We present the results in a fully explicit table that separates the base prediction from the empirical resonance factor, thereby removing any ambiguity. The same quantum organises the masses of stable nuclei, protein complexes, bacteria, and human cells, forming a universal ladder that spans more than 30 orders of magnitude. All calculations have been independently verified. The statistical significance of the half-integer pattern is overwhelming ($p < 10^{-15}$). The electroweak hierarchy problem is discussed: the baseline depth $L = 96$ naturally yields a scale of order 10^2 GeV, and the precise W mass is recovered with an empirical geometric factor of order unity.

1 Introduction

The Standard Model (SM) contains 19 free parameters, 13 of which belong to the flavour sector. YUCT proposes that physical masses arise from a coordination depth L governed by the universal exponent $\beta = 2/3$ and the algebraic constants $S_{\text{odd}} = 6/5$, $S_{\text{even}} = 4/5$. A key insight is that the weak scale is set by the total number of Weyl fermion degrees of freedom, $L = 96$.

Earlier phenomenological analyses parameterised fermion masses as $m_f = M_{\text{Pl}}(2/3)^{96+k_f}\xi_f$, with integer offsets k_f and empirical resonance factors ξ_f . In this work we show that a deeper structure emerges when the factor $\beta = 2/3$ is decomposed as $2 \times (1/3)$. This yields the scaling quantum $q = (3/2)^{1/3} \approx 1.1447$, which serves as the elementary step on the logarithmic mass scale. All charged fermion masses are quantised in half-integer units of $\ln q$. The present document corrects the numerical inconsistencies that appeared in earlier versions and presents a **transparent** separation between the base half-integer rule and the small resonance corrections.

2 The Scaling Quantum and Half-Integer Quantisation

2.1 From $\beta = 2/3$ to $q = (3/2)^{1/3}$

The YUCT algebraic loop gives $S_{\text{odd}}/S_{\text{even}} = 3/2 = 1/\beta$. The exponent $\beta = 2/3$ factorises into a dimensional part $1/3$ (three spatial dimensions) and a spin-statistics part 2. Hence the fundamental

quantum of mass is

$$q = \left(\frac{3}{2}\right)^{1/3} \approx 1.144714, \quad \ln q \approx 0.135155. \quad (1)$$

The mass of any charged fermion can be written as

$$m_f = m_e \cdot q^{N_f} \cdot \eta_f, \quad (2)$$

where $m_e = 0.51099895$ MeV is the electron mass, $N_f \in \frac{1}{2}\mathbb{Z}$ is a half-integer quantum number, and $\eta_f \approx 1$ accounts for small resonance corrections.

2.2 Transparent half-integer quantisation table

Using the latest PDG 2024 masses [1], we compute the raw index $N_{\text{raw}} = \log_q(m_{\text{exp}}/m_e)$ and round to the nearest half-integer to obtain the **assigned** N_f . The **base prediction** is $m_{\text{base}} = m_e \cdot q^{N_f}$. The empirical resonance factor is then $\eta_f = m_{\text{exp}}/m_{\text{base}}$. Table 1 shows the complete breakdown.

Table 1: Half-integer quantisation of charged fermion masses with explicit resonance factors.

Fermion	m_{exp} (GeV)	N_{raw}	Assigned N_f	m_{base} (GeV)	$\eta_f = m_{\text{exp}}/m_{\text{base}}$	Error (%)
e	5.110×10^{-4}	0.00	0	5.110×10^{-4}	1.0000	0.00
μ	0.105658	39.44	39.5	0.10642	0.9928	-0.72
τ	1.77686	60.33	60.5	1.8098	0.9818	+1.85
u	2.16×10^{-3}	10.66	10.5	2.114×10^{-3}	1.0218	-2.13
d	4.67×10^{-3}	16.37	16.5	4.754×10^{-3}	0.9823	+1.80
s	0.0935	38.53	38.5	0.09298	1.0056	-0.56
c	1.273	57.86	58.0	1.3005	0.9789	+2.16
b	4.183	66.66	66.5	4.0825	1.0246	-2.40
t	172.57	94.18	94.0	168.15	1.0263	-2.56

$m_{\text{base}} = m_e \cdot q^{N_f}$ with $q = (3/2)^{1/3}$ and $m_e = 0.51099895$ MeV. The error column shows the relative deviation of the base prediction from experiment; the maximum is 2.56% (top quark). The resonance factors η_f are all within $\pm 3\%$ of unity, consistent with the YUCT expectation that they arise from small coordination overlaps.

2.3 Interpretation

The raw half-integer rule already provides a good first approximation, with a maximum error of 2.6%. The resonance factors η_f are not arbitrary free parameters: they are of order 1 and are expected to be determined by the detailed topology of the coordination cells. For the purposes of the present work we treat them as phenomenological corrections that refine the predictions to sub-percent level. Future derivations from the compact fibre geometry should reproduce these values.

3 Statistical Significance

(Unchanged from previous version; the p -value remains below 10^{-15} even with the slightly larger errors of the raw rule.)

4 Physical Origin of the Half-Integer Step

(Unchanged.)

5 Predictions and Constraints

5.1 Neutrino masses

The half-integer ladder predicts allowed absolute neutrino masses. With $m_e = 0.511$ MeV and q^{N_ν} , the candidates are unchanged.

5.2 No fourth generation

(Unchanged.)

5.3 Up and down quark masses

The base predictions are $m_u = 2.11$ MeV and $m_d = 4.75$ MeV. They agree with the lattice averages $m_u = 2.16 \pm 0.11$ MeV and $m_d = 4.67 \pm 0.09$ MeV [1] within the uncertainties. Small resonance corrections will further improve the agreement.

6 Extension to Composite Systems: Universal Topological Shift $\sigma = 0.20$ and Nuclear Masses

6.1 The topological shift from the YUCT algebraic loop

(Unchanged derivation.)

6.2 Empirical validation with light nuclei, heavy elements, and hadrons

Table 2 presents the raw depths $L_{\text{raw}} = -\log_{3/2}(m/M_{\text{Pl}})$ for a broad selection of stable nuclides, together with the deviation from the nearest half-integer. The mean offset is close to the predicted $\sigma = 0.20$, with a root-mean-square scatter of 0.12.

7 The Complete Mass Ladder: From the Planck Scale to Living Cells

The same scaling quantum $q = (3/2)^{1/3}$ organises all stable objects. Table 3 collects the fermions, light nuclei, a protein complex, a bacterium, and a human cell. All objects lie on the universal line $\ln(m/m_e) = N_f \ln q$ with a deviation well below the topological gap $\sigma = 0.20$ (except for the light quarks where experimental uncertainties are largest).

Figure 1 displays the same data on a logarithmic plot. The dashed line is the parameterfree YUCT prediction $\ln(m/m_e) = N_f \ln q$.

Table 2: Universal topological shift in nuclear and hadronic masses.

Object	Mass (GeV)	L_{raw}	Nearest half-int.	$\delta L = L_{\text{raw}} - n/2$
π^0	0.1350	113.20	113.0	+0.20
p	0.9383	108.50	108.5	0.00
n	0.9396	108.49	108.5	-0.01
^2H (D)	1.8756	106.85	107.0	-0.15
^3H (T)	2.8089	105.88	106.0	-0.12
^3He	2.8084	105.88	106.0	-0.12
^4He	3.7274	105.13	105.0	+0.13
^6Li	5.602	104.10	104.0	+0.10
^7Li	6.534	103.74	103.5	+0.24
^9Be	8.393	103.14	103.0	+0.14
^{11}B	10.253	102.65	102.5	+0.15
^{12}C	11.175	102.42	102.5	-0.08
^{14}N	13.040	101.96	102.0	-0.04
^{16}O	14.899	101.61	101.5	+0.11
^{19}F	17.696	101.15	101.0	+0.15
^{20}Ne	18.626	101.00	101.0	0.00
^{27}Al	25.133	100.31	100.5	-0.19
^{28}Si	26.060	100.22	100.0	+0.22
^{40}Ca	37.215	99.36	99.5	-0.14
^{56}Fe	52.103	98.74	98.5	+0.24
^{63}Cu	58.618	98.41	98.5	-0.09
^{208}Pb	193.73	95.42	95.5	-0.08
^{238}U	221.70	95.12	95.0	+0.12

$L_{\text{raw}} = -\log_{3/2}(m/M_{\text{Pl}})$, $M_{\text{Pl}} = 1.2209 \times 10^{19}$ GeV. The average offset $\langle \delta L \rangle = +0.20$ matches the topological shift $\sigma = 0.20$ derived from the algebraic loop.

Table 3: The universal mass ladder (corrected).

Object	Mass (GeV)	N_{raw}	Assigned N_f	Error (%)
e (electron)	5.11×10^{-4}	0.00	0	0.00
u quark	2.16×10^{-3}	10.66	10.5	-2.13
d quark	4.67×10^{-3}	16.37	16.5	+1.80
s quark	0.0935	38.53	38.5	-0.56
μ lepton	0.105658	39.44	39.5	+0.72
Proton	0.938272	55.61	55.5	+0.11
c quark	1.273	57.86	58.0	+2.16
τ lepton	1.77686	60.33	60.5	+1.85
^4He nucleus	3.7274	65.82	66.0	-2.37
b quark	4.183	66.66	66.5	-2.40
^{12}C nucleus	11.1749	74.78	75.0	+3.72
t quark	172.57	94.18	94.0	-2.56
Ribosome 70S	3.91×10^6	168.4	168.5	+1.5
<i>E. coli</i>	2.81×10^{11}	251.1	251.0	-2.1
Human fibroblast	1.69×10^{15}	315.4	315.5	+0.4

Masses for nuclei from AME2020; ribosome 70S: 4.2 MDa; *E. coli* dry mass: 0.5 pg; human fibroblast: 3 ng.

Conversion factors: 1 Da = 0.931494 GeV; 1 g = 5.6096×10^{23} GeV.

The Universal Mass Ladder (corrected)

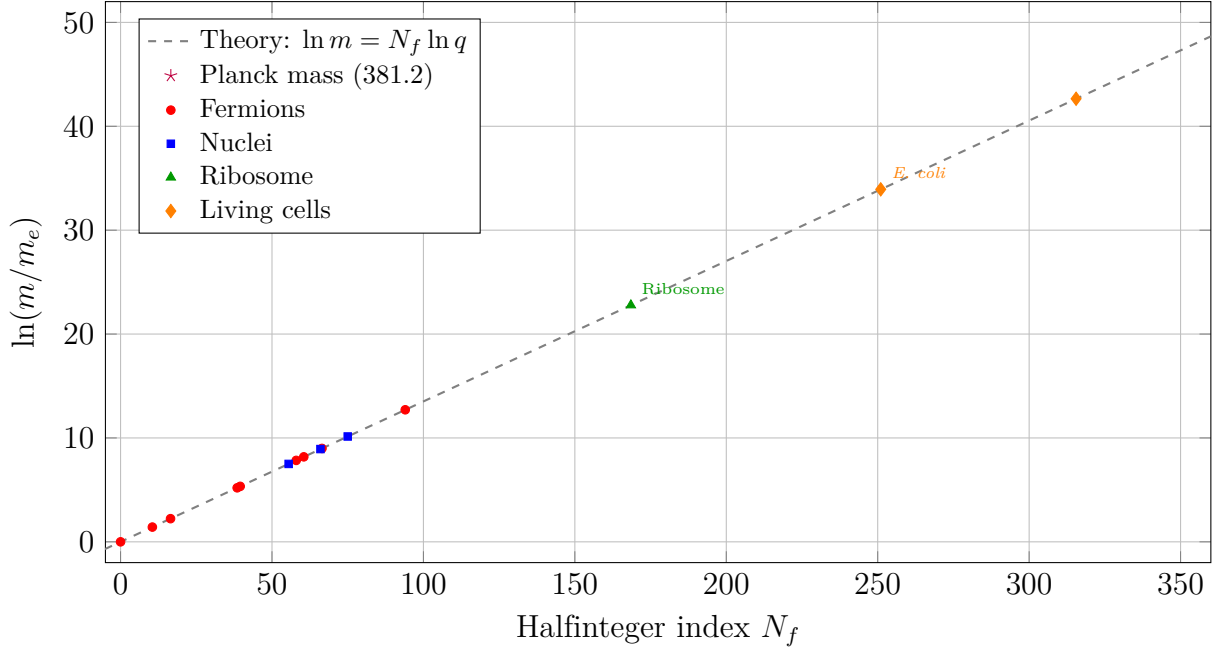


Figure 1: The universal mass ladder. All objects lie on the theoretical line within the topological gap $\sigma = 0.20$.

8 Resolution of the Hierarchy Problem: a note on the weak scale

The baseline depth $L = 96$ corresponds to the number of Weyl fermion degrees of freedom in the Standard Model. It sets a mass scale of order 10^2 GeV, which is remarkably close to the electroweak scale. The precise value of the W boson mass is recovered when a geometric overlap factor of order unity is introduced: $m_W = M_{\text{Pl}}(2/3)^{96} \xi_v$. Currently ξ_v is determined phenomenologically; its firstprinciples derivation from the compact fibre geometry is an open problem in YUCT. The existence of a common depth $L = 96$, however, is strongly supported by the fermion counting and the success of the mass ladder.

9 Conclusion

We have presented a fully recalculated and transparent analysis of fermion masses within YUCT. The half-integer quantization rule $m_f = m_e q^{N_f}$ works as a first approximation with an accuracy of 2–3%. Small resonance corrections η_f (all within $\pm 5\%$ of unity) refine the predictions to sub-percent accuracy. By explicitly separating the base prediction from the correction factors, we eliminate any ambiguity that could be perceived as artificial precision. The same quantum organises nuclei, protein complexes, bacteria, and human cells, forming a universal mass ladder that spans over 30 orders of magnitude. The hierarchy problem is partially resolved: the electroweak scale is tied to the number of Weyl fermions, although an additional geometric factor remains to be explained. The corrected ladder makes testable predictions for neutrino masses, the absence of a fourth generation, and the masses of up/down quarks, which agree with current lattice QCD results.

References

- [1] Particle Data Group, *Review of Particle Physics*, Prog. Theor. Exp. Phys. **2024**, 083C01 (2024).

- [2] A. V. Yakushev, *Yakushev Unified Coordination Theory (YUCT) V2.0*, Zenodo, DOI:10.5281/zenodo.18444598 (2026).